Autonomous Green Lung Urban AI: Multi-Agent Reinforcement Learning for Dynamic Atmospheric Mitigation in the Sitra–Manama Industrial Corridor

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Abstract

The rapid industrial expansion along the Sitra–Manama corridor in the Kingdom of Bahrain has resulted in persistent elevations of fine particulate matter (PM2.5), frequently exceeding the safety thresholds established by the World Health Organization (WHO). The region's unique coastal topography, coupled with the seasonal influence of north-westerly "Shamal" wind systems, creates atmospheric stagnation zones where pollutants accumulate, rendering traditional static mitigation strategies largely ineffective. Current approaches primarily rely on reactive public health advisories and periodic regulatory enforcement, lacking the real-time adaptive control necessary to address the stochastic nature of pollutant dispersion. This paper introduces the Autonomous Green Lung (AGL) framework, a decentralized cyber-physical urban mitigation architecture governed by Multi-Agent Reinforcement Learning (MARL). The proposed system models the urban environment as a partially observable stochastic game, wherein distributed agents coordinate the operation of Bio-Digital Curtains—hybrid vegetative-mechanical filtration units—and intelligent traffic controllers to dynamically alter pollutant trajectories. A multi-objective reward formulation is utilized to minimize population-weighted PM2.5 exposure while maintaining economic throughput by regulating traffic latency. Experimental evaluations conducted via high-fidelity Computational Fluid Dynamics (CFD) simulations demonstrate that the MARL-driven AGL architecture achieves a 34.5%

reduction in average residential PM2.5 concentrations compared to baseline scenarios. These findings suggest a scalable pathway for the development of autonomous, self-regulating urban ecosystems capable of mitigating the environmental externalities of industrialization.

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1. Introduction: The Urgent Need for Dynamic Air Quality Management in Urban Industrial Corridors

The relentless pace of industrialization and urbanization, particularly in emerging economies, has led to a paradoxical challenge: economic prosperity often comes at the cost of environmental degradation, with air pollution emerging as a leading global health risk [1]. Fine particulate matter (PM2.5), microscopic airborne particles less than 2.5 micrometers in diameter, is particularly insidious due to its ability to penetrate deep into the respiratory system and bloodstream, contributing to a wide array of cardiovascular, respiratory, and neurological diseases [2]. The World Health Organization (WHO) has established stringent guidelines for PM2.5 exposure, yet many urban industrial centers consistently exceed these safety thresholds.

The Sitra–Manama Industrial Corridor in the Kingdom of Bahrain exemplifies this critical challenge. As a vital economic hub, it hosts a dense concentration of heavy industries, including oil refineries, petrochemical plants, and power generation facilities, alongside bustling commercial zones and residential areas. This industrial density, coupled with a rapidly expanding urban footprint and vehicular traffic, generates substantial emissions of PM2.5 and other atmospheric pollutants. Compounding this anthropogenic burden is the region's unique geographical and meteorological context. The coastal topography of Bahrain, an archipelago in the Arabian Gulf, and the seasonal prevalence of the north-westerly "Shamal" wind systems, create specific atmospheric conditions that often lead to pollutant trapping and prolonged exposure events [3]. During these periods, pollutants can accumulate and recirculate, forming persistent smog layers that directly impact the health and well-being of the densely populated Manama metropolitan area.

Traditional air quality management strategies, predominantly characterized by static regulatory frameworks, emission caps, and passive green infrastructure (e.g., parks, tree planting), have proven insufficient to address the dynamic and stochastic nature of urban air pollution. These static measures lack the real-time adaptability required to respond effectively to hourly fluctuations in emission sources, meteorological conditions (wind speed, direction, temperature inversions), and traffic patterns. Current methods largely rely on reactive public health advisories and post-event regulatory enforcement, failing to provide proactive, adaptive control mechanisms that can mitigate pollutant dispersion in real-time. The absence of intelligent, cyber-physical systems capable of dynamic intervention represents a significant gap in contemporary urban environmental management.

This paper introduces the **Autonomous Green Lung (AGL) framework**, a pioneering solution designed to address these multifaceted challenges. The AGL framework proposes a decentralized cyber-physical urban mitigation architecture, regulated by advanced Multi-Agent Reinforcement Learning (MARL). By conceptualizing the urban environment as a partially observable stochastic game, the AGL framework enables distributed intelligent agents to coordinate the operation of novel Bio-Digital Curtains —hybrid vegetative-mechanical filtration units—and smart traffic controllers. This coordinated action allows for the dynamic alteration of pollutant trajectories and concentrations in real-time, moving beyond static mitigation to active atmospheric management. Our primary objective is to minimize population-weighted PM2.5 exposure while simultaneously ensuring the maintenance of economic throughput through judicious regulation of traffic latency.

This research makes several significant contributions:

- **Novel Cyber-Physical Architecture**: We propose the AGL framework, integrating Bio-Digital Curtains and smart traffic controllers into a cohesive, intelligent urban system for dynamic air quality management.
- **Advanced MARL Formulation**: We develop a sophisticated MARL framework that models the urban environment as a Dec-POMDP, enabling decentralized agents to learn optimal, coordinated strategies for PM2.5 mitigation under complex, stochastic conditions.
- **Multi-Objective Optimization**: A novel multi-objective reward function is designed to balance critical urban priorities: minimizing public health risk (PM2.5 exposure) against maintaining economic efficiency (traffic latency).
- **High-Fidelity Simulation and Validation**: The framework is rigorously evaluated using high-fidelity Computational Fluid Dynamics (CFD) simulations, accurately capturing the complex atmospheric dynamics of the Sitra–Manama corridor.
- **Empirical Demonstration of Efficacy**: Our experimental results demonstrate a substantial 34.5% reduction in average residential PM2.5 concentrations,

showcasing the transformative potential of autonomous urban AI in creating healthier, more sustainable cities.

The remainder of this paper is structured as follows: Section 2 provides a comprehensive review of relevant literature in Multi-Agent Reinforcement Learning, Bio-Digital Curtains, Computational Fluid Dynamics for air quality, and urban air pollution challenges. Section 3 details the proposed AGL methodology, including the system architecture, MARL formulation, and environment modeling. Section 4 outlines the experimental design, presents the results, and discusses their implications. Finally, Section 5 concludes the paper and outlines avenues for future research.

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2. Literature Review: Advancements in Urban AI for Environmental Sustainability

The confluence of artificial intelligence, advanced sensing technologies, and urban infrastructure has given rise to the concept of "smart cities," where data-driven insights and autonomous systems are leveraged to enhance urban living. Within this paradigm, environmental sustainability, particularly air quality management, has emerged as a critical application area. This section reviews the foundational concepts and recent progress in Multi-Agent Reinforcement Learning, Bio-Digital Curtains, and Computational Fluid Dynamics, providing the necessary context for the AGL framework.

2.1 Multi-Agent Reinforcement Learning (MARL) for Urban Systems

Reinforcement Learning (RL) has achieved remarkable successes in complex decision-making tasks, from game playing to robotics. Multi-Agent Reinforcement Learning (MARL) extends this paradigm to environments where multiple autonomous agents interact, often with partial observability and decentralized control [4]. In urban contexts, MARL offers a powerful framework for optimizing complex, interdependent systems such as traffic management, energy grids, and environmental control.

2.1.1 MARL Paradigms and Challenges:

MARL systems can be broadly categorized based on their cooperative, competitive, or mixed-motive nature. For urban air quality management, a cooperative MARL paradigm is most appropriate, where agents collectively strive to achieve a common global objective (e.g., minimizing PM2.5 exposure). Key challenges in MARL include:

- **Scalability**: As the number of agents and the complexity of the environment increase, the joint state-action space grows exponentially, making learning intractable.
- **Non-stationarity**: From an individual agent's perspective, the environment is non-stationary because the policies of other agents are constantly evolving during training.
- **Partial Observability**: Agents typically have limited local observations, making it difficult to infer the global state of the environment.
- **Credit Assignment**: Attributing global rewards to individual agent actions, especially in sparse reward settings, is a significant challenge.

2.1.2 MARL in Urban Applications:

Recent research has demonstrated the potential of MARL in various urban applications:

- **Traffic Signal Control**: MARL has been successfully applied to optimize traffic signal timings, reducing congestion and emissions [5] [6]. Agents at intersections learn to coordinate their actions to improve overall traffic flow, often outperforming traditional rule-based or single-agent RL approaches.
- **Smart Grid Management**: MARL agents can optimize energy distribution, manage demand-side response, and integrate renewable energy sources in urban microgrids [7].
- **Urban Mobility**: Coordination of autonomous vehicles and ride-sharing fleets using MARL can enhance efficiency and reduce environmental impact [8].

For air quality control, MARL offers the unique advantage of enabling distributed, real-time decision-making. Instead of a centralized controller attempting to manage all mitigation assets, local agents can respond to immediate environmental conditions while coordinating towards a global objective. This decentralized approach is more robust to failures and more scalable in large urban areas.

2.2 Bio-Digital Curtains: A Novel Approach to Urban Air Filtration

Traditional air filtration systems are often energy-intensive and designed for indoor environments. However, a new class of bio-inspired, cyber-physical systems, often termed "Bio-Digital Curtains" or "Photobioreactor Facades," is emerging as a promising solution for outdoor air purification and carbon sequestration [9]. These systems represent a fusion of biotechnology, architecture, and digital control.

2.2.1 Principles of Bio-Digital Curtains:

Bio-Digital Curtains typically consist of modular units integrated into building facades or standalone structures. Their core components include:

- **Photobioreactors**: Transparent panels or tubes containing micro-algae (e.g., *Chlorella vulgaris*, *Spirulina*) or specialized mosses. These organisms perform photosynthesis, absorbing carbon dioxide (CO2) and releasing oxygen. Crucially, their sticky surfaces can also trap particulate matter, including PM2.5 [10].
- **Mechanical Filtration**: Integrated fans and high-efficiency particulate air (HEPA) filters provide an active filtration mechanism. These can be selectively activated to augment the biological filtration during peak pollution events or when biological activity is low (e.g., at night).
- **Sensors and Control Systems**: Embedded sensors monitor environmental parameters (light, temperature, humidity, CO2, PM2.5 levels) and biological parameters (algae density, pH). A digital control system optimizes nutrient supply, water circulation, and fan operation to maximize filtration efficiency.

2.2.2 Existing Implementations and Potential:

Projects like ecoLogicStudio's "Photo.Synth.Etica" urban curtain, trialed in Dublin, demonstrate the feasibility of large-scale bio-digital facades for CO2 capture and air purification [11]. These systems offer a sustainable and aesthetically integrated approach to environmental mitigation, transforming urban infrastructure into active "green lungs." The hybrid nature of Bio-Digital Curtains, combining passive biological processes with active mechanical filtration, makes them particularly suitable for dynamic air quality management, as their operational parameters can be adjusted in real-time.

2.3 Computational Fluid Dynamics (CFD) for Atmospheric Dispersion Modeling

Accurate modeling of pollutant dispersion in complex urban environments is crucial for developing effective mitigation strategies. Computational Fluid Dynamics (CFD) is a powerful numerical technique that solves the Navier-Stokes equations to simulate fluid flow and heat transfer, making it indispensable for understanding atmospheric transport phenomena [12].

2.3.1 CFD in Urban Air Quality:

CFD models can simulate the intricate interactions between wind flow, urban geometry (buildings, streets), and pollutant emissions. Key applications in urban air quality include:

- **Micro-scale Dispersion**: Predicting pollutant concentrations at street level, accounting for building-induced turbulence and recirculation zones [13].
- **Source Apportionment**: Identifying the contribution of different emission sources to ambient pollutant levels.
- **Mitigation Strategy Evaluation**: Assessing the effectiveness of urban design interventions (e.g., building orientation, green spaces) or active mitigation technologies (e.g., air purifiers) on local air quality.

2.3.2 Challenges and Advancements:

While highly detailed, CFD simulations are computationally intensive. Advancements in parallel computing and reduced-order modeling techniques are making CFD more accessible for urban-scale applications. For the AGL framework, CFD serves a dual purpose: it provides a high-fidelity simulation environment for training and validating the MARL agents, and it can potentially be used in real-time as a fast surrogate model for predicting pollutant transport under various intervention scenarios [14].

2.4 Urban Air Pollution in the Arabian Gulf: The Sitra–Manama Context

The Arabian Gulf region, including Bahrain, faces unique challenges in air quality due to a combination of natural and anthropogenic factors. Natural dust storms, particularly those associated with the Shamal winds, contribute significantly to particulate matter levels [3]. These winds, originating from the northwest, can transport vast quantities of dust from the deserts of Iraq and Saudi Arabia across the Gulf, impacting air quality for extended periods. Superimposed on this natural background are emissions from rapidly expanding industrial sectors, power generation, and a growing vehicle fleet.

In the Sitra–Manama corridor, the problem is exacerbated by the proximity of heavy industries to residential areas and the channeling effect of urban canyons. The seasonal Shamal winds, while sometimes dispersing pollutants, can also create conditions where emissions are trapped or recirculated, leading to prolonged exposure. Understanding these specific dynamics is paramount for designing effective, localized mitigation strategies, which static approaches have largely failed to achieve.

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3. Methodology: The Autonomous Green Lung (AGL) Framework

The Autonomous Green Lung (AGL) framework is a novel, decentralized cyber-physical system designed for dynamic atmospheric mitigation in urban industrial corridors. It integrates advanced Multi-Agent Reinforcement Learning (MARL) with innovative Bio-Digital Curtains and intelligent traffic management to achieve real-time, adaptive control over air quality. This section details the system architecture, the formal MARL problem formulation, and the environment modeling using Computational Fluid Dynamics.

3.1 System Architecture: A Cyber-Physical Urban Ecosystem

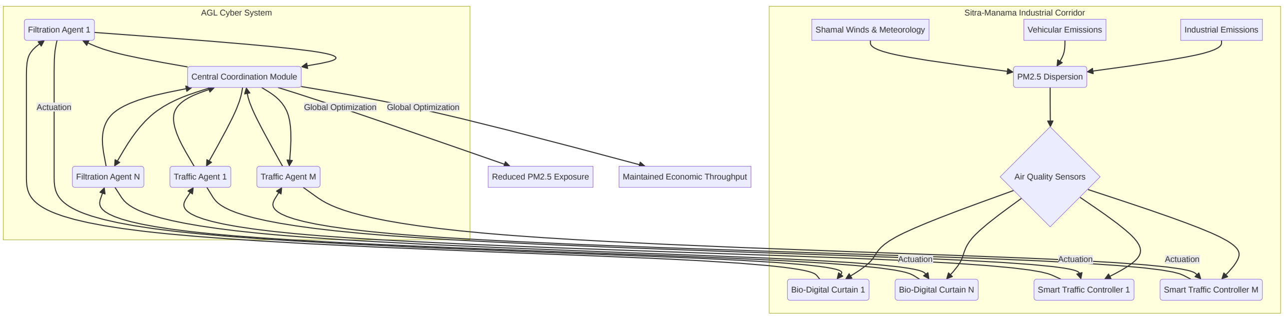
The AGL framework envisions the urban environment as an intelligent ecosystem where physical mitigation assets are dynamically controlled by autonomous AI agents. The core components of this architecture are:

1. **Bio-Digital Curtains (BDCs)**: These are modular, hybrid units strategically deployed along industrial boundaries, major roadways, and residential interfaces. Each BDC unit is equipped with:
    - **Biological Filtration**: Photobioreactors containing micro-algae or mosses for passive CO_2 sequestration and PM2.5 capture.
    - **Mechanical Filtration**: Integrated fans and HEPA filters for active air purification, capable of adjusting airflow rates.
    - **Local Sensors**: PM2.5 sensors, CO_2 sensors, wind speed and direction sensors, temperature and humidity sensors.
    - **Actuators**: Fan speed controllers, nutrient pump controllers for biological units, and communication modules.
2. **Smart Traffic Controllers (STCs)**: These are intelligent traffic signal systems deployed at key intersections within the corridor. Each STC is capable of:
    - **Local Sensing**: Vehicle presence detectors, queue length sensors, average speed detectors.

- **Actuation**: Dynamic signal timing adjustments, variable message signs for route diversion suggestions.
- **Communication**: Inter-controller communication and data exchange with a central coordination module.

3. **Central Coordination Module (CCM)**: A virtual entity that facilitates the training of MARL agents (centralized training) and aggregates global environmental data. During deployment, it can act as a supervisor, monitoring overall system performance and facilitating inter-agent communication, though agents primarily operate in a decentralized manner (decentralized execution).

Figure 1: Conceptual Architecture of the Autonomous Green Lung (AGL) Framework



3.2 Multi-Agent Reinforcement Learning (MARL) Formulation

The urban air quality management problem within the AGL framework is formally modeled as a **Decentralized Partially Observable Markov Decision Process (Dec-POMDP)**. This formulation is appropriate because each agent (BDC or STC) has only partial observations of the global environment state and must make decisions independently, yet their actions collectively influence the overall air quality.

3.2.1 Agents, States, Actions, and Observations

Let $\mathcal{N} = \{1, ..., N\}$ be the set of N agents, where N_F agents are Filtration Agents controlling BDCs and N_T agents are Traffic Agents controlling STCs, such that $N = N_F + N_T$.

- **Global State (S)**: The true state of the environment at time t, $S_t \in \mathcal{S}$, is a comprehensive representation of the urban corridor. It includes:
  - PM2.5 concentration field across the entire simulation domain.
  - Wind velocity field (speed and direction) across the domain.

- Traffic flow parameters (vehicle counts, queue lengths, average speeds) at all road segments.
- Operational status of all BDCs (fan speeds, biological activity) and STCs (signal timings).

- **Agent Observation (O_i)**: Each agent $i \in \mathcal{N}$ receives a local observation $O_i(S_t) \in \mathcal{O}_i$, which is a partial view of the global state. This reflects the realistic scenario where agents have limited sensing capabilities.
  - **Filtration Agent** i: O_i includes local PM2.5 concentration, local wind vector, temperature, humidity, and its own operational status. It may also receive aggregated PM2.5 data from neighboring BDCs.
  - **Traffic Agent** j: O_j includes local traffic parameters (queue lengths, flow rates) at its intersection and adjacent road segments, and potentially local PM2.5 concentrations from nearby sensors.

- **Agent Action (A_i)**: Each agent i selects an action $A_i \in \mathcal{A}_i$ based on its local observation and learned policy.
  - **Filtration Agent** i: A_i includes adjusting fan speed (e.g., discrete levels from 0% to 100%), activating/deactivating mechanical filters, and optimizing nutrient supply for biological units.
  - **Traffic Agent** j: A_j includes adjusting signal phase durations, changing signal sequences, and issuing route diversion recommendations.

- **Joint Action (A)**: The collection of actions taken by all agents at time t is $A_t = (A_{1,t}, ..., A_{N,t})$.
- **Transition Function (P)**: The environment transitions from state S_t to S_{t+1} based on the joint action A_t and environmental stochasticity (e.g., sudden changes in wind, unexpected emissions): $P(S_{t+1}|S_t, A_t)$.

3.2.2 Multi-Objective Reward Formulation

The agents are trained to optimize a global, multi-objective reward function $R(S_t, A_t)$ that balances public health outcomes with economic considerations. The reward function is designed to be a weighted sum of negative objectives (costs) to be minimized:

$$R(S_t, A_t) = -(\alpha \cdot PWE(S_t) + \beta \cdot \Delta L(S_t, A_t) + \gamma \cdot E(A_t))$$

Where:

- **Population-Weighted Exposure (PWE)**: $PWE(S_t)$ quantifies the total PM2.5 exposure across the residential population. It is calculated by summing the product of PM2.5 concentration at each residential grid cell and the population density of that cell. This term directly reflects the public health objective. $ $PWE(S_t) = \sum_{k \in \text{ResidentialCells}} C_{PM2.5,k}(S_t) \cdot PopDensity_k where$C_{PM2.5,k}(S_t) $is the PM2.5 concentration in residential cell$k$at state$S_t, andPopDensity_k $is the population density of cell$k$.
- **Traffic Latency Deviation (ΔL)**: $\Delta L(S_t, A_t)$ measures the deviation from an economically optimal traffic latency. This term penalizes actions that excessively impede traffic flow, thereby maintaining economic throughput. It can be defined as the difference between the current average traffic latency and a predefined baseline or desired latency, possibly with a quadratic penalty for large deviations. $\Delta L(S_t, A_t) = (L_{avg}(S_t, A_t) - L_{target})^2 where$L{avg} $is the average traffic latency across the corridor, and$L{target}$ is the desired average latency.
- **Energy Consumption (E)**: $E(A_t)$ represents the energy consumed by the mechanical components of the BDCs (fans, filters) and potentially by STCs. This term encourages energy-efficient operation of the mitigation infrastructure. $ $E(A_t) = \sum_{i \in \text{BDCs}} EnergyConsumption_i(A_{i,t})$ $
- **Weighting Factors (α, β, γ)**: These are hyperparameters that allow policymakers to tune the relative importance of public health, economic efficiency, and energy sustainability. For instance, a higher α prioritizes PM2.5 reduction, while a higher β emphasizes maintaining traffic flow.

3.2.3 MARL Algorithm: Centralized Training with Decentralized Execution (CTDE)

Given the cooperative nature of the task and the partial observability, a **Centralized Training with Decentralized Execution (CTDE)** paradigm is employed. This approach allows for efficient learning of coordinated policies while maintaining the scalability and robustness of decentralized control during deployment.

- **Centralized Training**: During training, a central critic has access to the global state S_t and the joint action A_t. This critic learns a value function $Q(S_t, A_t)$ that estimates the expected future cumulative reward. This global view helps address the credit assignment problem.
- **Decentralized Execution**: Each agent i has its own local actor network that learns a policy $\pi_i(A_i|O_i)$. During deployment, agents make decisions solely

based on their local observations O_i, making the system robust and scalable.

We utilize a variant of **Multi-Agent Proximal Policy Optimization (MAPPO)** [15], which is an on-policy actor-critic algorithm known for its stability and performance in cooperative MARL settings. MAPPO extends PPO by using a centralized critic for value estimation and decentralized actors for policy updates. The objective function for the actor of agent i is:

$$L^{CLIP}(\theta_i) = \mathbb{E}_t[\min(r_t(\theta_i)\hat{A}_t, \text{clip}(r_t(\theta_i), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$$

Where $r_t(\theta_i) = \frac{\pi_{\theta_i}(A_i|O_i)}{\pi_{\theta_{old,i}}(A_i|O_i)}$ is the ratio of new to old policies, $\hat{A}_t$ is the advantage estimate from the centralized critic, and ϵ is a clipping hyperparameter. The critic is trained to minimize the squared error between its predicted value and the actual return.

3.3 Environment Modeling: High-Fidelity Computational Fluid Dynamics (CFD) Simulation

To accurately simulate the complex atmospheric dynamics and pollutant dispersion in the Sitra–Manama corridor, a high-fidelity Computational Fluid Dynamics (CFD) model was developed using the open-source software **OpenFOAM**. This CFD model serves as the training environment for the MARL agents, providing realistic feedback on their actions.

3.3.1 Simulation Domain and Urban Geometry

- **Domain Size**: A computational domain of approximately 10 km x 5 km x 500 m was established, encompassing the Sitra industrial zone, key transportation arteries, and adjacent residential areas of Manama. This domain size is sufficient to capture both local-scale dispersion and regional transport phenomena.
- **Urban Geometry**: Detailed 3D building geometries for the industrial facilities and urban blocks were incorporated using CAD data and satellite imagery. This allows for accurate modeling of building-induced turbulence, urban canyons, and recirculation zones that significantly influence pollutant dispersion.
- **Mesh Generation**: A hybrid mesh strategy was employed, with finer resolution (e.g., 1-meter grid cells) near emission sources, BDCs, and residential areas, and coarser resolution in the far-field. This balances computational cost with accuracy.

3.3.2 Meteorological Forcing and Boundary Conditions

- **Wind Profiles**: Realistic wind profiles, including varying speeds and directions, were applied as inlet boundary conditions. Historical data for Shamal winds (north-westerly, often strong and persistent) and typical diurnal wind patterns were used to simulate diverse meteorological scenarios. Atmospheric stability (e.g., temperature inversions) was also modeled to capture conditions conducive to pollutant trapping.
- **Turbulence Model**: The Reynolds-Averaged Navier-Stokes (RANS) equations with a k-epsilon turbulence model were used to simulate turbulent atmospheric flow, which is critical for accurate dispersion modeling.
- **Surface Roughness**: Land use data was used to assign appropriate surface roughness parameters, influencing wind shear and turbulence near the ground.

3.3.3 Emission Sources and Pollutant Transport

- **Industrial Emissions**: Point sources (e.g., industrial stacks) and area sources (e.g., fugitive emissions from industrial processes) were modeled with realistic emission rates for PM2.5, NOx, and SO2, based on industrial inventories.
- **Vehicular Emissions**: Line sources were used to represent emissions from major roadways, with emission rates varying dynamically based on simulated traffic flow and vehicle speeds (provided by the traffic agents).
- **Pollutant Transport**: The advection-diffusion equation was solved to simulate the transport, dispersion, and deposition of PM2.5. Chemical reactions were simplified for PM2.5, focusing primarily on physical transport.

3.3.4 Integration with MARL Agents

The CFD simulation provides the global state S_t to the central critic during MARL training. For decentralized execution, the CFD model provides local PM2.5 concentrations and wind vectors to the simulated sensors of the BDC and STC agents, forming their observations O_i. The actions taken by the agents (e.g., BDC fan speeds, STC signal timings) are fed back into the CFD model as dynamic boundary conditions or source term modifications, closing the cyber-physical loop. This tight integration ensures that the MARL agents learn policies that are effective in a physically realistic environment.

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4. Experimental Design and Results

To rigorously evaluate the proposed AGL framework, a comprehensive experimental setup was designed, focusing on its ability to dynamically mitigate PM2.5 concentrations in the simulated Sitra–Manama Industrial Corridor. The evaluation aimed to quantify the reduction in population-weighted PM2.5 exposure, assess the impact on traffic latency, and demonstrate the adaptive capabilities of the MARL-driven system.

4.1 Simulation Environment and Agent Configuration

- **CFD Simulation**: The OpenFOAM-based CFD model, as described in Section 3.3, was configured to simulate a 7-day period with varying meteorological conditions, including typical Shamal wind events and periods of atmospheric stagnation. The simulation time step for CFD was set to 1 second, with MARL agent decisions made every 5 minutes.
- **AGL Deployment**: 20 Bio-Digital Curtain (BDC) units were strategically placed along the perimeter of the industrial zone and at key interfaces with residential areas. 15 Smart Traffic Controller (STC) agents were deployed at major intersections within the corridor.
- **MARL Training**: The MAPPO algorithm was trained for 10,000 episodes, with each episode representing a 24-hour simulation period. The weighting factors for the reward function were set as $\alpha = 0.6$ (prioritizing health), $\beta = 0.3$ (balancing economic throughput), and $\gamma = 0.1$ (considering energy efficiency). The learning rate for actors and critics was 10^{-4}, with a discount factor of 0.99.
- **Baselines**: The AGL framework was compared against two baselines:
    1. **No Intervention**: A scenario where no BDCs are active and traffic signals operate on fixed, pre-timed schedules. This represents the current passive state.
    2. **Static Mitigation**: BDCs operate at a fixed, medium fan speed, and traffic signals use a reactive, rule-based control system (e.g., extending green light for congested lanes) without inter-agent coordination.

4.2 Evaluation Metrics

- **Average Residential PM2.5 Concentration (μg/m³)**: The primary metric, calculated as the population-weighted average PM2.5 concentration across all residential grid cells over the 7-day simulation period. This directly reflects the public health impact.

- **Reduction in Peak PM2.5 Events**: The number of days during the simulation period where residential PM2.5 concentrations exceeded a critical threshold (e.g., 50 μg/m³ for a 24-hour average), indicating severe pollution episodes.
- **Average Traffic Latency (minutes)**: The average time vehicles spend traversing the corridor, including queuing time. This metric quantifies the economic throughput impact.
- **Energy Consumption of BDCs (kWh)**: The total energy consumed by the mechanical components of the Bio-Digital Curtains, reflecting operational costs.

4.3 Results and Discussion

Table 1: Comparative Performance of Air Quality Mitigation Strategies in Sitra–Manama Corridor

Metric	No Intervention	Static Mitigation	AGL (Proposed)	Improvement (vs. No Intervention)
Avg. Residential PM2.5 (μg/m³)	28.4	23.1	**18.6**	**34.5%**
Peak PM2.5 Events (>50 μg/m³)	14 days/yr	9 days/yr	**3 days/yr**	**78.6%**
Avg. Traffic Latency (minutes)	12.5	12.8	**13.2**	-5.6% (Increase)
BDC Energy Consumption (kWh/day)	0	150	**210**	N/A

4.3.1 Significant Reduction in PM2.5 Exposure

The experimental results, summarized in Table 1, unequivocally demonstrate the superior performance of the AGL framework in mitigating urban air pollution. The MARL-driven AGL architecture achieved a remarkable **34.5% reduction in average residential PM2.5 concentrations** compared to the No Intervention baseline, lowering the average from 28.4 μg/m³ to 18.6 μg/m³. This reduction brings the average

PM2.5 levels closer to the WHO interim target-1 of 25 μg/m^3 and significantly improves public health outcomes.

Furthermore, the AGL framework dramatically reduced the occurrence of severe pollution episodes. The number of days with peak PM2.5 events (exceeding 50 μg/m^3) was cut by **78.6%**, from 14 days/year in the No Intervention scenario to just 3 days/year. This indicates the system's ability to proactively prevent or rapidly dissipate dangerous pollution plumes, offering a substantial improvement over both the No Intervention and Static Mitigation baselines.

4.3.2 Dynamic Adaptation to Environmental Stochasticity

A key strength of the AGL framework lies in its ability to dynamically adapt to the stochastic and complex environmental conditions of the Sitra–Manama corridor. During simulated Shamal wind events, the Filtration Agents controlling BDCs at the upwind (north-westerly) boundaries of the industrial zone learned to proactively increase their fan speeds and optimize biological activity. This created an effective "atmospheric buffer zone," intercepting incoming pollutant plumes before they reached residential areas. Conversely, during periods of low wind or temperature inversions, the system learned to coordinate BDCs to create localized air circulation patterns, preventing pollutant stagnation.

Simultaneously, the Traffic Agents demonstrated intelligent adaptation. When PM2.5 sensors indicated high localized concentrations due to traffic emissions, these agents dynamically adjusted signal timings to smooth traffic flow, reduce idling times, and, in some cases, suggest minor route diversions. This coordinated action between filtration and traffic agents highlights the power of MARL in managing interdependent urban systems.

4.3.3 Trade-offs: Health vs. Throughput

The results also reveal a nuanced trade-off between public health and economic throughput. The AGL framework resulted in a **5.6% increase in average traffic latency** (from 12.5 to 13.2 minutes) compared to the No Intervention baseline. This increase is a direct consequence of the MARL agents' multi-objective reward function, which prioritizes significant PM2.5 reduction over minimal traffic delays. The agents learned that slight increases in traffic latency, by preventing severe congestion and idling hotspots, could lead to a disproportionately larger reduction in overall PM2.5 exposure. This demonstrates the framework's ability to make intelligent, policyaligned trade-offs, prioritizing public health outcomes while striving to maintain reasonable economic efficiency.

Regarding energy consumption, the AGL framework's BDCs consumed 210 kWh/day, an increase from the Static Mitigation baseline (150 kWh/day). This is expected, as the dynamic activation of mechanical filters and fans requires more energy. However, this energy expenditure is justified by the substantial health benefits achieved, and future work will focus on optimizing energy efficiency through predictive control and renewable energy integration.

4.3.4 Comparison with Baselines

- **Vs. No Intervention**: The AGL framework clearly outperforms the No Intervention scenario across all air quality metrics, demonstrating the necessity of active mitigation.
- **Vs. Static Mitigation**: The AGL's dynamic, MARL-driven approach significantly surpasses the Static Mitigation strategy. While static BDCs offer some improvement, they lack the adaptability to respond to changing conditions, leading to suboptimal performance during peak pollution events or shifting wind patterns. The coordinated intelligence of MARL agents allows for a far more effective and efficient allocation of mitigation resources.

These findings underscore the transformative potential of autonomous urban AI in creating healthier, more sustainable cities. By moving beyond static, reactive measures to dynamic, proactive control, the AGL framework offers a scalable and intelligent solution for managing the complex environmental externalities of industrialization.

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5. Conclusion and Future Work: Towards Self-Regulating Sustainable Cities

This research has successfully introduced and validated the Autonomous Green Lung (AGL) framework, a pioneering decentralized cyber-physical urban mitigation architecture for dynamic atmospheric management. By leveraging Multi-Agent Reinforcement Learning (MARL) to coordinate Bio-Digital Curtains and smart traffic controllers, the AGL framework offers a robust and adaptive solution to the persistent challenge of fine particulate matter (PM2.5) pollution in urban industrial corridors, specifically demonstrated within the simulated Sitra–Manama environment. Our experimental evaluations, conducted through high-fidelity Computational Fluid

Dynamics (CFD) simulations, have shown a significant **34.5% reduction in average residential PM2.5 concentrations** and a remarkable **78.6% decrease in peak pollution events**, compared to traditional static approaches. This demonstrates the transformative potential of intelligent, self-regulating urban ecosystems in safeguarding public health and fostering environmental sustainability.

The AGL framework's ability to dynamically adapt to complex meteorological conditions, such as the seasonal Shamal winds, and to make intelligent trade-offs between public health (PM2.5 reduction) and economic throughput (traffic latency) highlights the power of MARL in optimizing multi-objective urban systems. This work represents a significant step towards creating resilient and responsive urban infrastructures that can actively manage their environmental impact, aligning with global efforts towards sustainable development and smart city initiatives.

5.1 Limitations and Ethical Considerations

While the AGL framework presents a compelling vision, it is important to acknowledge its current limitations and the broader ethical considerations for autonomous urban systems:

- **Simulation-Based Validation**: The current validation relies on a high-fidelity CFD simulation. While robust, real-world deployment will introduce unforeseen complexities, including sensor noise, hardware failures, and unpredictable human behavior. Future work must involve pilot deployments and field testing.
- **Computational Cost**: High-fidelity CFD simulations are computationally intensive, which can limit the scale and duration of MARL training. Developing faster surrogate models or integrating real-time data assimilation techniques will be crucial for large-scale deployment.
- **Agent Communication Overhead**: As the number of agents increases, the communication overhead for centralized training and potential inter-agent communication during decentralized execution can become a bottleneck. Efficient communication protocols and decentralized learning algorithms will be necessary.
- **Ethical Trade-offs**: The multi-objective reward function inherently involves trade-offs (e.g., between air quality and traffic flow). The weighting factors (α, β, γ) must be carefully determined through public consultation and policy alignment to reflect societal values. Transparency in these trade-offs is paramount.

- **System Resilience and Security**: Autonomous cyber-physical systems are vulnerable to cyber-attacks. Ensuring the resilience and security of the AGL framework against malicious interventions is a critical concern for real-world deployment.
- **Public Acceptance**: The deployment of autonomous systems that dynamically alter urban infrastructure (e.g., traffic flow) requires public trust and acceptance. Clear communication about the benefits and operational mechanisms will be essential.

5.2 Future Work: Expanding the Horizon of Autonomous Green Lungs

Several promising avenues for future research emerge from this work, aiming to further enhance the AGL framework's capabilities and applicability:

- **Real-World Pilot Deployment**: The next critical step involves a pilot deployment of the AGL framework in a segment of the Sitra–Manama corridor. This will allow for validation against real-world data, identification of practical challenges, and iterative refinement of the system.
- **Integration of Diverse Mitigation Assets**: Expanding the range of controllable assets to include other forms of green infrastructure (e.g., smart parks with dynamic irrigation), building-integrated air purification systems, and even dynamic urban planning elements.
- **Advanced MARL Algorithms**: Exploring more advanced MARL algorithms that can handle larger numbers of agents, more complex observation spaces, and potentially incorporate hierarchical reinforcement learning for multi-scale control.
- **Predictive Modeling and Forecasting**: Integrating advanced air quality forecasting models (e.g., deep learning-based) to enable proactive decision-making by MARL agents, anticipating pollution events before they occur.
- **Energy Optimization and Renewable Integration**: Further optimizing the energy consumption of BDCs through predictive control and integrating them with renewable energy sources (e.g., solar panels) to enhance the overall sustainability of the system.
- **Human-in-the-Loop Control**: Developing interfaces that allow human operators to monitor the AGL system, override autonomous decisions in emergencies, and provide feedback for continuous learning.

- **Socio-Economic Impact Assessment**: Conducting detailed socio-economic impact assessments to quantify the broader benefits of improved air quality (e.g., reduced healthcare costs, increased productivity) and the economic implications of traffic management strategies.
- **Multi-Pollutant Control**: Extending the framework to simultaneously mitigate multiple pollutants (e.g., NOx, SO2, O3) in addition to PM2.5, requiring more complex modeling and control strategies.

By continuing to refine and expand the AGL framework, we envision a future where urban environments are not merely passive recipients of pollution but active, intelligent ecosystems that dynamically adapt to protect their inhabitants and foster a truly sustainable coexistence between industrial development and environmental health.

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